



Midnight Relay Tour

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Difficulty Level : Medium-Hard

Midnight Relay Tour

Minutes before the GCPC live stream goes on air, a thunderstorm knocks several relay stations offline across the city. Nadia starts at the backup control hub in city 1, and the final broadcast center is in city n .

There are k damaged relays, each located in a different city. Nadia must visit every one of those cities at least once to reboot the relay manually. After repairing all of them, she must continue to city n . The road network is bidirectional, every road has a travel time, and Nadia may pass through the same city or road multiple times if that leads to a faster overall route.

Task: Compute the minimum total travel time needed to start at city 1, visit all damaged relay cities in any order, and finish at city n . If this is impossible, print -1.

Input

The first line contains three integers n , m , and k :

- n — the number of cities
- m — the number of roads
- k — the number of damaged relays

The second line contains k distinct integers $c_1, c_2, \dots, c_k$ — the cities containing damaged relays.

Each of the next m lines contains three integers u , v , and w , describing a bidirectional road between cities u and v with travel time w .

Output

Print a single integer:

- the minimum total travel time, or
- -1 if Nadia cannot visit all damaged relays and still reach city n .

Constraints

- $2 \leq n \leq 10^5$
- $0 \leq m \leq 2 \cdot 10^5$
- $1 \leq k \leq 15$
- $2 \leq c_i \leq n - 1$

- All relay cities are distinct
- $1 \leq u, v \leq n, u \neq v$
- $1 \leq w \leq 10^9$
- Roads are bidirectional
- Multiple roads between the same pair of cities may exist
- Time limit: **3 seconds**
- Memory limit: **512 MB**

Sample Input 1

```
6 9 2
3 5
1 2 4
1 3 2
2 3 1
2 4 7
3 4 3
3 5 8
4 5 2
4 6 5
5 6 1
```

Sample Output 1

```
8
```

Sample Input 2

```
5 3 2
2 4
1 2 3
2 3 4
4 5 1
```

Sample Output 2

```
-1
```

Explanation

In the first sample, an optimal route is:

$$1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$$

with total travel time

$$2 + 3 + 2 + 1 = 8.$$

This route visits both damaged relays (3 and 5) before reaching city 6.

In the second sample, the graph is split into two disconnected components, so it is impossible to visit relay city 4 after starting from city 1.